Name: Key Period: _____ Date: _____

## Topic 4 Practice Test: 4.0, 4.2–4.6 — KEY

*Show all your work. Indicate clearly the methods you use, because you will be graded on the correctness of your methods as well as the accuracy of your final answers. No calculator.*

**Multiple Choice (#1–5). Identify the choice that best completes the statement or answers the question.**

**1.** C If $\Delta MNP \cong \Delta QRS$, then $\overline{PM} \cong$

**A** $\overline{QR}$ **B** $\overline{RS}$ **C** $\overline{SQ}$ **D** $\overline{NP}$

**2.** C Is the following statement true: A right triangle can also be an obtuse triangle.

**A** Always True **B** Sometimes True **C** Never True **D** Can't be determined

**3.** B Which is a correct classification of $\Delta ABC$ if $m\angle A = 38°$ and $m\angle B = 52°$?

**A** Acute **B** Right **C** Obtuse **D** Equiangular

**4.** B Is the following statement true: An isosceles triangle is also an equilateral triangle.

**A** Always True **B** Sometimes True **C** Never True **D** Can't be determined

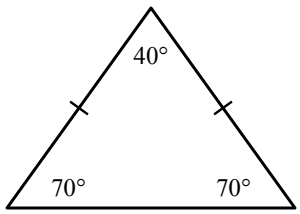
**5.** C Which of the following can <u>not</u> be used to prove two triangles are congruent?

**A** SAS **B** AAB **C** SSA **D** HL

**Classifying Triangles (#6–11).**

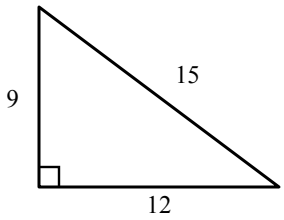
**6.** Classify each triangle by its angles *and* sides.

**a.**



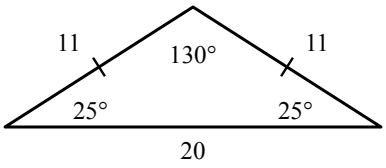

acute isosceles

**b.**



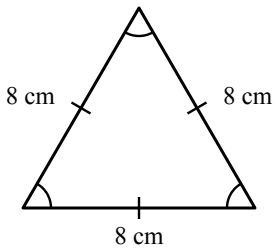

right scalene

**c.**



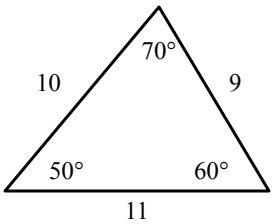

obtuse isosceles

**d.**



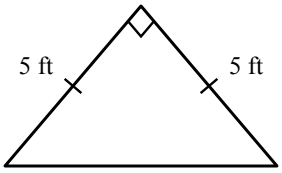

equiangular equilateral

**e.**




acute scalene

**f.**




right isosceles

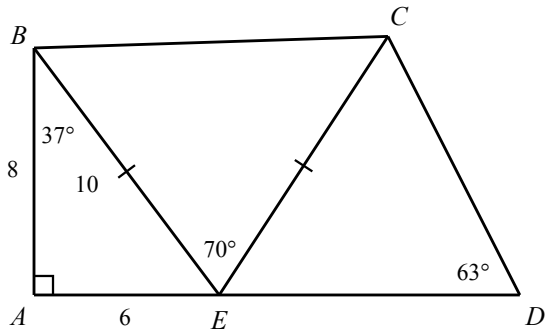
**7.** Points $A$, $E$, and $D$ are on one line. Classify each triangle by its angles and sides. *(Find the missing angles first.)*

a. $\Delta ABE$: right scalene

b. $\Delta BEC$: acute isosceles

c. $\Delta CED$: acute scalene

d. $m\angle ECD =$ 60°




$m\angle AEB = 180 - 90 - 37 = 53° \rightarrow m\angle CED = 180 - 53 - 70 = 57°$ (straight line)
$m\angle ECD = 180 - 57 - 63 = 60° \rightarrow$ angles 57, 60, 63 all different = scalene. $\Delta BEC$
base angles = 55°, 55°

**8.** Use the distance formula to find each side length, then classify.

**a.** $A(1, 2)$, $B(4, 6)$, $C(8, 3)$. Classify by sides and angles.

$AB = \sqrt{((4-1)^2 + (6-2)^2)} = \sqrt{25} = 5$
$BC = \sqrt{((8-4)^2 + (3-6)^2)} = \sqrt{25} = 5$
$AC = \sqrt{((8-1)^2 + (3-2)^2)} = \sqrt{50} = 5\sqrt{2}$
2 sides $\cong \rightarrow$ isosceles
$5^2 + 5^2 = 50 = (\sqrt{50})^2 \rightarrow$ right $\angle B$
right isosceles

**b.** $P(0, 0)$, $Q(6, 0)$, $R(2, 5)$. Classify by sides.

$PQ = \sqrt{((6-0)^2 + 0^2)} = 6$
$QR = \sqrt{((2-6)^2 + (5-0)^2)} = \sqrt{41}$
$PR = \sqrt{((2-0)^2 + (5-0)^2)} = \sqrt{29}$
all 3 different
scalene

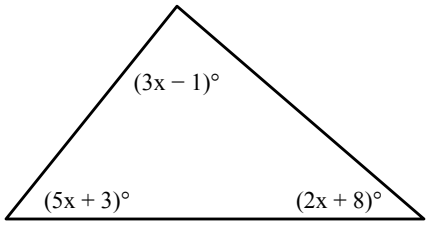
**9.** In $\Delta ABC$, the measure of $\angle B$ is 12 more than twice the measure of $\angle A$, and the measure of $\angle C$ is 12 less than three times the measure of $\angle A$. Find the measure of each angle, then classify $\Delta ABC$ by its angles and sides.

$x + 2x + 12 + 3x - 12 = 180 \rightarrow 6x = 180 \rightarrow x = 30$
$m\angle A = 30°, m\angle B = 72°, m\angle C = 78°$ 3 diff angles = 3 diff sides acute scalene

**10.** In a right triangle, the measure of one acute angle is 18° less than twice the measure of the other acute angle. Find the measures of the acute angles.

$x + (2x - 18) + 90 = 180 \rightarrow 3x = 108 \rightarrow x = 36$ angles: 36°, 54°

**11.** Find the value of $x$. Then classify the triangle by its angles and sides.




$5x + 3 + 3x - 1 + 2x + 8 = 180$

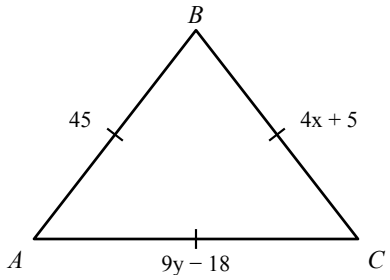
$10x + 10 = 180 \rightarrow \boxed{x = 17}$

angles: $88°, 50°, 42°$

**acute scalene**

**Isosceles and Equilateral Triangles (#12–21).**

**12.** If $\Delta ABC$ is an equilateral triangle, solve for both $x$ and $y$.




$4x + 5 = 45 \rightarrow 4x = 40 \rightarrow \boxed{x = 10}$

$9y - 18 = 45 \rightarrow 9y = 63 \rightarrow \boxed{y = 7}$

**13.** $\Delta STU$ is an equilateral triangle. If $ST$ is 3 less than twice $x$, $TU$ is 6 more than $x$, and $SU$ is 21 less than four times $x$, find $x$ and the measure of each side.

$2x - 3 = x + 6 \rightarrow \boxed{x = 9}$ check: $4(9) - 21 = 15$ ✓

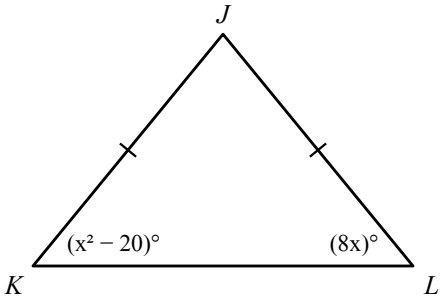
**ST = TU = SU = 15 units**

**14.** $\Delta KLM$ is an isosceles triangle with $\overline{KL} \cong \overline{KM}$. If $KL = 5x - 8$, $KM = 3x + 6$, and $LM = 2x + 3$, find $x$, the measure of each side, and the perimeter.

$5x - 8 = 3x + 6 \rightarrow 2x = 14 \rightarrow \boxed{x = 7}$

$KL = KM = 27, LM = 17$ perimeter $= 27 + 27 + 17 = \boxed{71 \text{ units}}$

**15.** In $\Delta JKL$, $\overline{JK} \cong \overline{JL}$. Solve for $x$. Then find $m\angle J$ and classify $\Delta JKL$ by its angles and sides.




base $\angle$'s $\cong$: $x^2 - 20 = 8x$

$x^2 - 8x - 20 = 0$

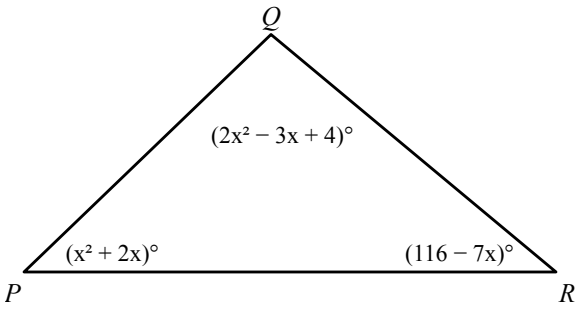
$(x - 10)(x + 2) = 0$

$\boxed{x = 10}$ or $x = -2 \leftarrow 8(-2) = -16°$ not possible

base $\angle$'s $= 80°$ $m\angle J = 180 - 160 = \boxed{20°}$

**acute isosceles**

**16.** Using the integer value of $x$, classify $\Delta PQR$ by its angles and sides.




$x^2 + 2x + 2x^2 - 3x + 4 + 116 - 7x = 180$

$3x^2 - 8x - 60 = 0$

$(3x + 10)(x - 6) = 0$

$x = -10/3$ or $\boxed{x = 6}$ ← integer value

$m\angle P = 48°$, $m\angle Q = 58°$, $m\angle R = 74°$

**acute scalene**

**17.** Find and use the value of $x$ that makes sense to find $FI$ and $IN$. Then classify $\Delta FIN$ by its angles and sides.

$FI = (2x - 3)^2$

$IN = 6(3 - 2x)$

$FN = 36$

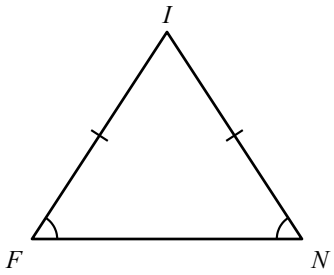



*Figure not drawn to scale.*

$(2x - 3)^2 = 6(3 - 2x)$

$4x^2 - 12x + 9 = 18 - 12x$

$4x^2 - 9 = 0$ ← diff of squares

$(2x - 3)(2x + 3) = 0$

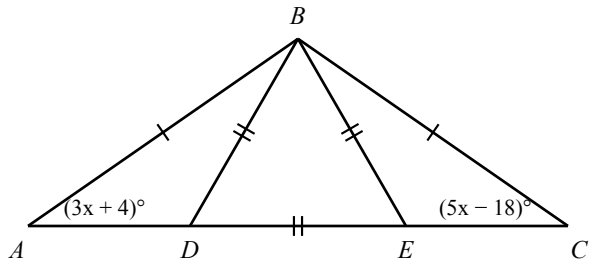
$x = 3/2$ ← $FI = 0$ does not work

$\boxed{x = -3/2}$

$\boxed{FI = 36, IN = 36}$ all sides ≅ →

**equilateral & equiangular**

**18.** If $\Delta ABC$ is isosceles with $\overline{AB} \cong \overline{CB}$ and $\Delta DBE$ is an equilateral triangle, find each value. Points $A$, $D$, $E$, $C$ are on one line.




$x =$ 11

$m\angle A =$ 37°

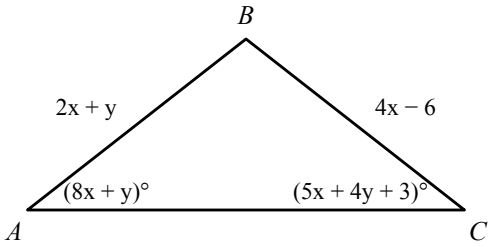
$m\angle ADB =$ 120°

$m\angle ABD =$ 23°

$m\angle ABC =$ 106°

$3x + 4 = 5x - 18$ → $x = 11$ → $m\angle A = 37°$. $\angle ADB$ & $\angle BDE$ linear pair: $180 - 60 = 120°$. $180 - 37 - 120 = 23°$. $23 + 60 + 23 = 106°$

**19.** $\Delta ABC$ is isosceles with vertex $B$. Solve for $x$ and $y$. Then find $m\angle B$ and classify $\Delta ABC$ by its angles. *Hint: you will need a system of equations.*




sides: $2x + y = 4x - 6 \rightarrow y = 2x - 6$

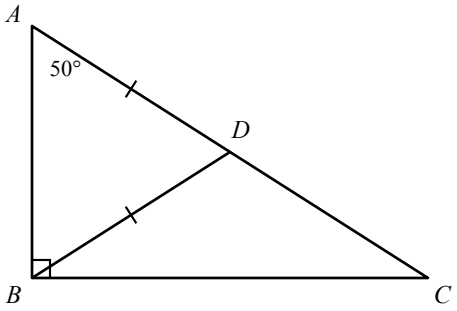
base $\angle$'s: $8x + y = 5x + 4y + 3 \rightarrow 3x - 3y = 3 \rightarrow x - y = 1$

substitute: $x - (2x - 6) = 1 \rightarrow -x = -5 \rightarrow$ **$x = 5$**

$y = 2(5) - 6 \rightarrow$ **$y = 4$**

$m\angle A = m\angle C = 44° \rightarrow m\angle B = 180 - 88 =$ **$92°$** **obtuse isosceles**

**20.** In right $\Delta ABC$, $D$ is on $\overline{AC}$ and $\overline{AD} \cong \overline{BD}$. If $m\angle A = 50°$, find each measure and classify $\Delta BDC$.




a. $m\angle ABD =$ **$50°$**

b. $m\angle DBC =$ **$40°$**

c. $m\angle BDC =$ **$100°$**

d. $\Delta BDC$: **obtuse isosceles**

$\Delta ADB$ isosceles $\rightarrow 50°$. $90 - 50 = 40°$.

$m\angle C = 40° \rightarrow 180 - 40 - 40 = 100°$

**21.** **ERROR ANALYSIS** In $\Delta XYZ$, $\overline{XY} \cong \overline{XZ}$ and $m\angle X = 40°$. Describe and correct the error below.

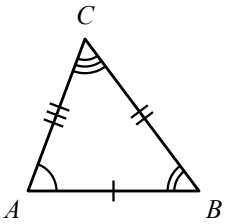
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Base angles are congruent, so
m∠Y = m∠X = 40°
m∠Z = 180 - 40 - 40 = 100°
```

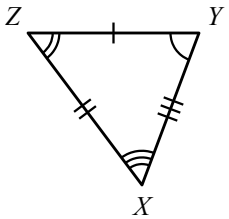
The base $\angle$'s are opposite the $\cong$ sides, so $\angle Y \cong \angle Z$, not $\angle X$. $\angle X$ is the vertex angle.

$(180 - 40) \div 2 = 70 \rightarrow$ **$m\angle Y = m\angle Z = 70°$**

**Congruence Statements and CPCTC (#22–31).**

**22.** Write three valid congruency statements for the triangles below.




a. **$\Delta ABC \cong \Delta YZX$**

b. **$\Delta BCA \cong \Delta ZXY$**

c. **$\Delta CAB \cong \Delta XYZ$**

**23.** Given $\Delta JKL \cong \Delta PQR$, complete each of the following statements.

a) $\angle K \cong$ **$\angle Q$** b) $\overline{LJ} \cong$ **$RP$** c) $\Delta LKJ \cong$ **$\Delta RQP$**

d) **$KL$** $\cong \overline{QR}$ e) $\angle R \cong$ **$\angle L$** f) $\Delta QPR \cong$ **$\Delta KJL$**

**24.** Given $\Delta DGK \cong \Delta RTW$, list all congruent angles and sides.

| Angles | Sides |
|---|---|
| $\angle D \cong \angle R$ | $DG \cong RT$ |
| $\angle G \cong \angle T$ | $GK \cong TW$ |
| $\angle K \cong \angle W$ | $DK \cong RW$ |

**25.** $\Delta FGH \cong \Delta LMN$. $FG = 12$ cm, $GH = 9$ cm, $NL = 17$ cm, $m\angle F = 34°$, and $m\angle H = 105°$. Find each missing measure. *Hint: draw the triangles if needed.*

a) $LM =$ 12 cm  b) $MN =$ 9 cm  c) $FH =$ 17 cm

d) $m\angle L =$ 34°  e) $m\angle N =$ 105°  f) $m\angle M =$ 41°

$m\angle G = 180 - 34 - 105 = 41° \rightarrow \angle M \cong \angle G$ by CPCTC

**26.** $\Delta ABC \cong \Delta DEF$. $AB = 3x + 4$, $DE = 5x - 10$, $BC = x + y$, $m\angle C = (4y + 6)°$, and $m\angle F = 70°$. Find $x$, $y$, and $EF$.

$3x + 4 = 5x - 10 \rightarrow 14 = 2x \rightarrow \boxed{x = 7}$  $4y + 6 = 70 \rightarrow 4y = 64 \rightarrow \boxed{y = 16}$

$EF = BC = 7 + 16 = \boxed{23}$

**27.** $\Delta XYZ \cong \Delta UVW$. $m\angle X = 47°$, $m\angle W = 90°$, $m\angle Y = (6y + 1)°$, $m\angle V = (4z - 1)°$, $XZ = 3x - 5$, and $UW = 2x + 9$. Find $x$, $y$, and $z$.

$\angle Z \cong \angle W = 90° \rightarrow m\angle Y = 180 - 47 - 90 = 43°$

$6y + 1 = 43 \rightarrow \boxed{y = 7}$  $4z - 1 = 43 \rightarrow \boxed{z = 11}$  $3x - 5 = 2x + 9 \rightarrow \boxed{x = 14}$

**28.** $\Delta JKL \cong \Delta MNO$. Solve for $x$ and $y$, then find $NO$.

$JK = 3x + y$,  $JL = 4x - y$,  $KL = 2x + 3y$,  $MN = 18$,  $MO = 17$

$3x + y = 18$

$4x - y = 17$ (add) $\rightarrow 7x = 35 \rightarrow \boxed{x = 5}$  $3(5) + y = 18 \rightarrow \boxed{y = 3}$

$NO = KL = 2(5) + 3(3) = \boxed{19}$

**29.** $\Delta ABC \cong \Delta XYZ$. $AB = 2x - 4$, $BC = x + 9$, $AC = 3x - 10$, and $XY = 18$. Find the perimeter of $\Delta ABC$.

$2x - 4 = 18 \rightarrow x = 11$  $BC = 20$, $AC = 23$  perimeter $= 18 + 20 + 23 = \boxed{61 \text{ units}}$

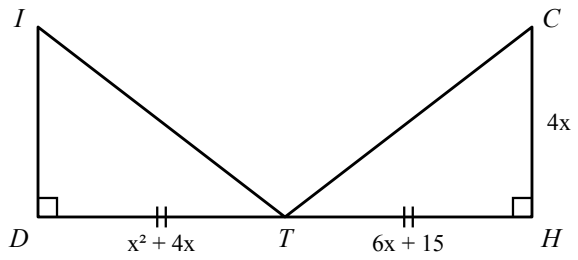
**30.** $\Delta GHI \cong \Delta MNO$. $GH = x^2 - 3x$, $MN = 2x + 14$, and $GI = 5x + 6$. Solve for $x$, then find $GH$. Check every solution.

$x^2 - 3x = 2x + 14 \rightarrow x^2 - 5x - 14 = 0 \rightarrow (x - 7)(x + 2) = 0$

$\boxed{x = 7}$ or $x = -2 \leftarrow GI = 5(-2) + 6 = -4$ not possible

$GH = 49 - 21 = \boxed{28}$

**31.** $\Delta TID \cong \Delta TCH$. Solve for $x$. Then find the area of $\Delta TID$.




$DT = HT: x^2 + 4x = 6x + 15$

$x^2 - 2x - 15 = 0 \rightarrow (x - 5)(x + 3) = 0$

$\boxed{x = 5}$ ($x = -3 \rightarrow CH = -12$ not possible)

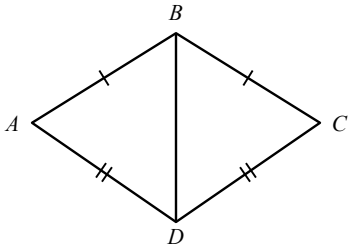
$DT = 45$, $ID = CH = 4(5) = 20$

area $= \frac{1}{2}(45)(20) = \boxed{450\ u^2}$

**Triangle Congruence Shortcuts: SSS, SAS, ASA, AAS, HL (#32–42).**

**32.** Complete the triangle congruence statement and tell which shortcut supports it. If the triangles cannot be shown to be congruent, write "CBD" (cannot be determined) and leave the statement blank.

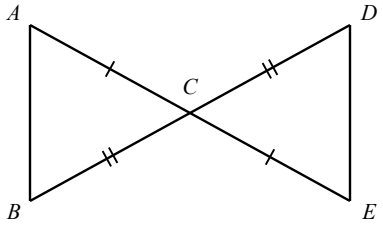
**a.**




$\Delta ABD \cong$ $\Delta CBD$ Reason: SSS

*shared side

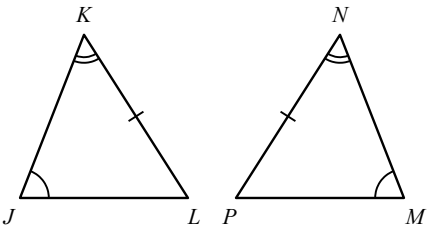
**b.**




$\Delta ABC \cong$ $\Delta EDC$ Reason: SAS

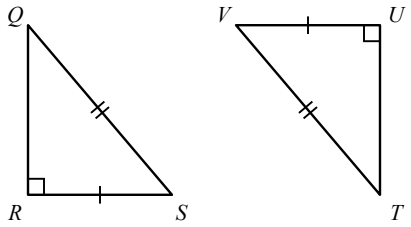
*vertical ∠'s

**c.**



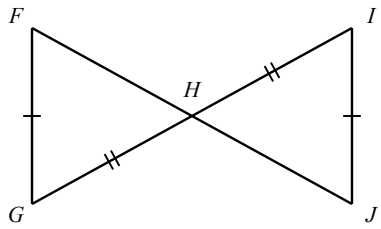

$\Delta JKL \cong$ $\Delta MNP$ Reason: AAS

**d.**



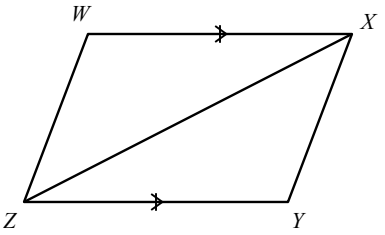

$\Delta QRS \cong$ $\Delta TUV$ Reason: HL

**e.**




$\Delta FGH \cong$ ✕ Reason: CBD *SSA

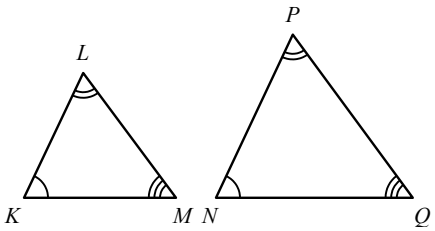
**f.**




$\Delta WXZ \cong$ $\Delta YZX$ Reason: SAS *AIA

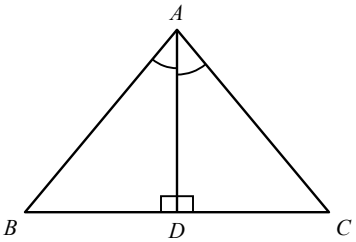
+ shared

**g.**




$\Delta KLM \cong$ ✕ Reason: CBD *AAA

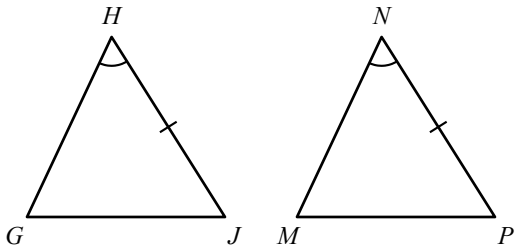
**h.**




$\Delta ABD \cong$ $\Delta ACD$ Reason: ASA

*shared side

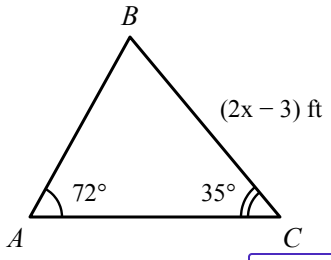
**33.** What additional piece of information is needed to show $\Delta GHJ \cong \Delta MNP$ by each shortcut?

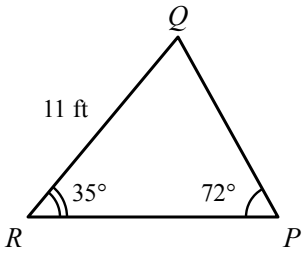



a) SAS: GH ≅ MN  b) ASA: ∠J ≅ ∠P  c) AAS: ∠G ≅ ∠M

**34.** For what value of $x$ would each pair of triangles be congruent by the given shortcut?

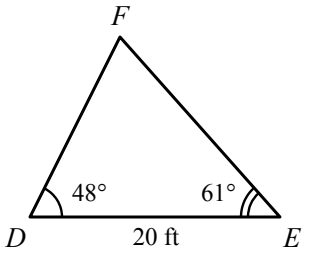
**a. AAS**

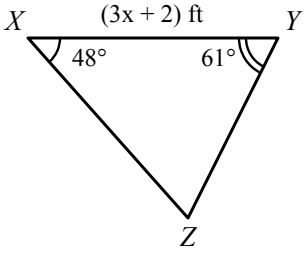



2x − 3 = 11 → **x = 7**

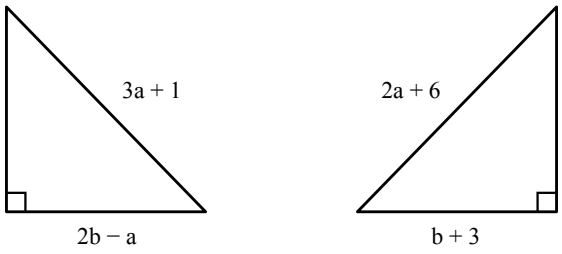
**b. ASA**




3x + 2 = 20 → **x = 6**

**35.** For what values of $a$ and $b$ are the triangles congruent by the HL Theorem? The labeled legs correspond.



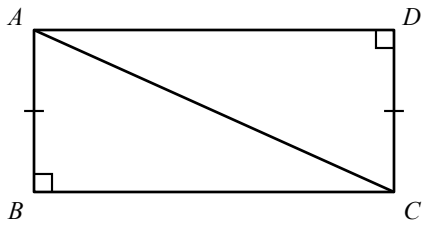

hyp: 3a + 1 = 2a + 6 → **a = 5**

legs: 2b − a = b + 3

2b − 5 = b + 3 → **b = 8**

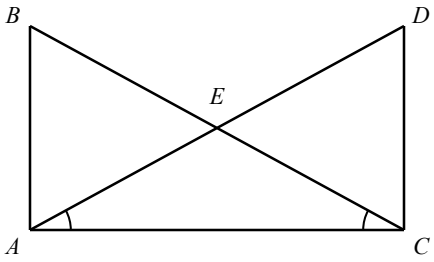
(hyp = 16, leg = 11 ✓)

**36.** **Understand** Is there sufficient information to prove $\Delta ABC \cong \Delta CDA$ by SSS? Explain.




Yes. AC is shared and AB ≅ CD. ∠B and ∠D are right angles, so the Pythagorean Theorem gives BC = DA. That's 3 pairs of ≅ sides → SSS.

**37.** $\overline{AD}$ and $\overline{CB}$ intersect at $E$, and $\angle EAC \cong \angle ECA$.




**a.** Why is $\overline{AE} \cong \overline{CE}$?

ΔAEC has 2 ≅ angles, so the sides opposite them are ≅ (isosceles)

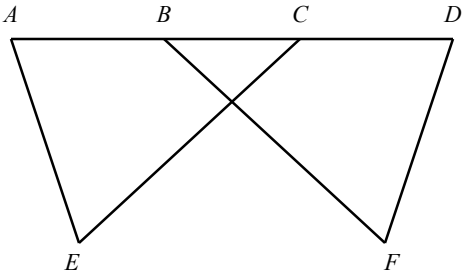
**b.** What additional statement do you need to prove $\Delta ABE \cong \Delta CDE$ by SAS? BE ≅ DE

**c.** What additional statement do you need to prove $\Delta ABE \cong \Delta CDE$ by AAS? ∠B ≅ ∠D

**d.** Suppose $\Delta ABE \cong \Delta CDE$. Explain why $\Delta ABC \cong \Delta CDA$ and name the shortcut.

AB ≅ CD and ∠BAE ≅ ∠DCE by CPCTC. Add ∠EAC ≅ ∠ECA → ∠BAC ≅ ∠DCA (angle addition).
AC is shared → ΔABC ≅ ΔCDA by SAS

**38.** Points $A$, $B$, $C$, $D$ are on one line and $\Delta ACE \cong \Delta DBF$. Identify which side or angle is congruent to each given part.




a. $\angle ACE \cong$ ∠DBF

b. $\overline{CE} \cong$ BF

c. $\angle AEC \cong$ ∠DFB

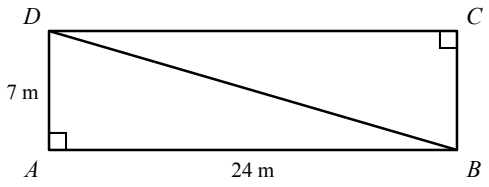
**d.** Explain why $\overline{AB} \cong \overline{CD}$.

AC ≅ DB by CPCTC. Both contain BC (shared), so AB = AC − BC = DB − BC = CD

**39.** **Apply** A triangular piece for a phone stand has sides of 9 cm and 12 cm with a 50° angle between them. The slot it goes into is a triangle with sides of 12 cm and 9 cm with a 50° angle between them. Will the piece fit exactly in the slot? Explain.

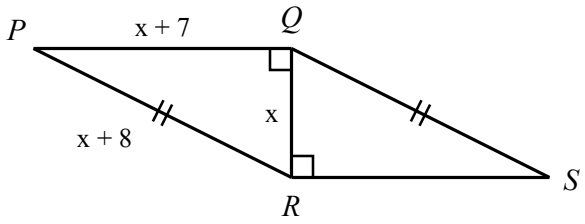
Yes. The 50° angle is between the 9 and 12 cm sides in both, so the triangles are ≅ by SAS. Same size and shape → it fits.

**40.** A drone flies the path $A \to B \to D \to C$ at a constant speed of 4 meters per second. In the figure, $\Delta ABD \cong \Delta CDB$. How long does the trip take? Round to the nearest tenth of a second.



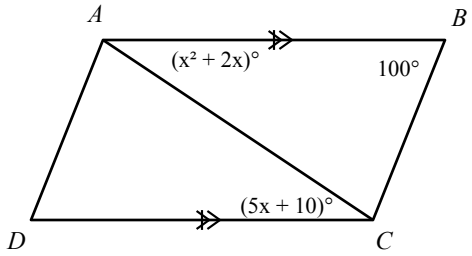

$BD = \sqrt{(24^2 + 7^2)} = \sqrt{625} = 25$ m (pyth. thm)
$DC \cong BA$ by CPCTC → 24 m
total = 24 + 25 + 24 = 73 m
$d = rt \to 73 = 4t \to$ **$t \approx 18.3$ seconds**

**41.** Solve for $x$. Explain why $\Delta PQR \cong \Delta SRQ$, then find the perimeter of $\Delta SRQ$.




pyth. thm: $x^2 + (x + 7)^2 = (x + 8)^2$
$x^2 + x^2 + 14x + 49 = x^2 + 16x + 64$
$x^2 - 2x - 15 = 0 \to (x - 5)(x + 3) = 0$
**$x = 5$** (no negative side lengths)
HL: right ∠'s, hyp $PR \cong SQ$, $QR$ shared
perimeter = 5 + 12 + 13 = **30 units**

**42.** $\overline{AB} \parallel \overline{DC}$ and $\overline{AB} \cong \overline{DC}$. Solve for $x$, then find each measure.




a. $x =$ 5
b. $m\angle BAC =$ 35°
c. $m\angle BCA =$ 45°
d. $\Delta ABC \cong \Delta CDA$ by SAS
e. $m\angle D =$ 100°

AIA: $x^2 + 2x = 5x + 10 \to x^2 - 3x - 10 = 0 \to (x - 5)(x + 2) = 0 \to x = 5$ ($x = -2$ gives 0°)
$m\angle BAC = 35°$ $m\angle BCA = 180 - 100 - 35 = 45°$ $\angle D \cong \angle B$ by CPCTC

**Challenge (#43–45). These take several steps. Check every solution you get from factoring.**

**43.** Solve for $x$ and $y$. Then find all three angle measures and classify the triangle by its angles and sides.

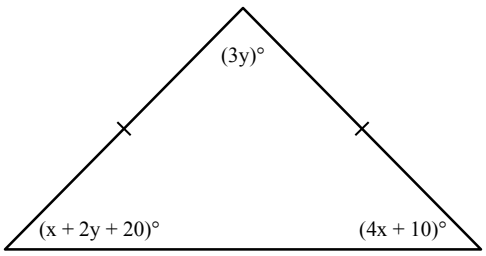



*Figure not drawn to scale.*

base ∠'s: $x + 2y + 20 = 4x + 10 \to 2y = 3x - 10$
sum: $3y + x + 2y + 20 + 4x + 10 = 180$
$5x + 5y = 150 \to y = 30 - x$
$2(30 - x) = 3x - 10 \to 70 = 5x \to$
**$x = 14$**, **$y = 16$**
angles: **48°, 66°, 66°** **acute isosceles**

**44.** $\Delta ABC$ has vertices $A(-4, 1)$, $B(-1, 5)$, $C(-1, 1)$. $\Delta DEF$ has vertices $D(2, -1)$, $E(2, -4)$, $F(6, -4)$.

a. Prove the triangles are congruent. Write a congruence statement with the vertices in the correct order and name the shortcut.

b. Find $m\angle A + m\angle F$ without a protractor. Explain.

c. Classify $\Delta DEF$ by its angles and sides, and find its area.

$AB = \sqrt{(3^2 + 4^2)} = 5$ $BC = \sqrt{(0^2 + 4^2)} = 4$ $AC = \sqrt{(3^2 + 0^2)} = 3$

$DE = \sqrt{(0^2 + 3^2)} = 3$ $EF = \sqrt{(4^2 + 0^2)} = 4$ $DF = \sqrt{(4^2 + 3^2)} = 5$

a. $A$ is between the 5 & 3 sides → $D$; $B$ is between 5 & 4 → $F$; $C \to E$

$\boxed{\Delta ABC \cong \Delta DFE \text{ by SSS}}$

b. $\angle F \cong \angle B$ by CPCTC. $\angle C$ is a right angle, so $\angle A + \angle B = 90 \to \boxed{90^\circ}$

c. right angle at $E$, 3 different sides → $\boxed{\text{right scalene}}$ area = $\frac{1}{2}(3)(4) = \boxed{6 \text{ u}^2}$

**45.** $\Delta ACB \cong \Delta DCE$. Points $A$, $C$, $D$ are on one line, and points $B$, $C$, $E$ are on one line. Find $x$ and $y$. Then find $m\angle B$ and classify $\Delta DCE$ by its angles and sides.

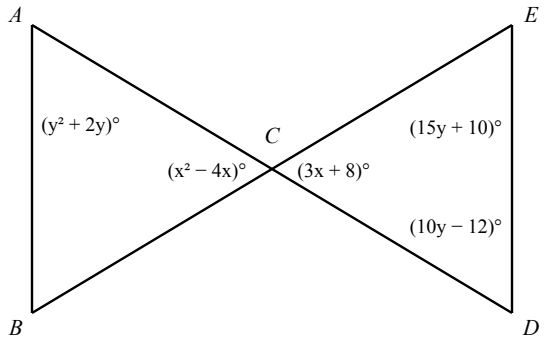



*Figure not drawn to scale.*

VA: $x^2 - 4x = 3x + 8 \to (x - 8)(x + 1) = 0 \to x = 8$ or $-1$

CPCTC $\angle A \cong \angle D$: $y^2 + 2y = 10y - 12 \to (y - 2)(y - 6) = 0 \to y = 2$ or $6$

all roots give positive $\angle$'s → use $\Delta$ sum in $\Delta DCE$:

$(3x + 8) + (10y - 12) + (15y + 10) = 180$

only $x = 8$, $y = 6$ works: $32 + 48 + 100 = 180$ ✓

$\boxed{x = 8, y = 6}$ $\angle B \cong \angle E \to$ $\boxed{m\angle B = 100^\circ}$

$\boxed{\text{obtuse scalene}}$